# 졸업프로젝트 (시작반)

Course Introduction

# 1학기 일정 - 수업 1

- 내용: 요구사항 분석서 작성법, 연구 진행 방법

# 1학기 일정 - 제안서/요구사항 분석서

- 양식: 제안서

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2. 프로젝트 목표
   2.1 목표
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# 1학기 일정 - 제안서/요구사항 분석서

- 양식: 요구사항 분석서 1차

1 개요

1.1 프로젝트 기획 배경

1.2 기술 동향

1.3 프로젝트 주요 기능 및 특징

1.4 조원 구성 및 역할 분담

1.5 일정

2 기능적 요구사항

2.1 Top Level Use Case Diagram
(또는 Top level 기능 설명)

2.2 각 기능별 동작 시나리오
(또는 Use Case Document)

3 비기능적 요구사항 (아래 사항 중 해당 프로젝트와 관련된 것만 기술)

3.1 사용편리성 (예제 포함)

3.2 신뢰성

3.3 성능

3.4 이식성

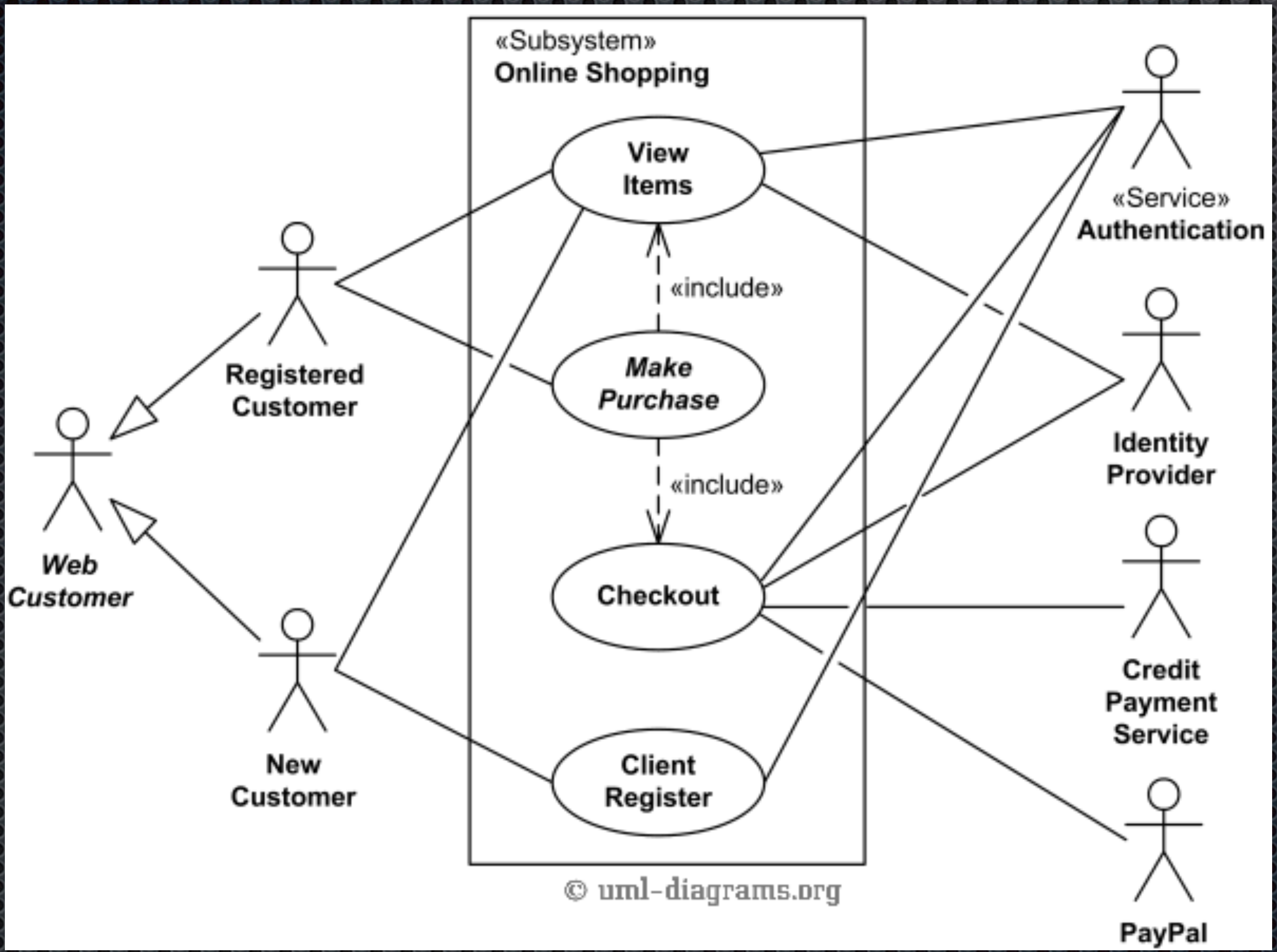
3.5 유지관리

3.6 구현상 제약사항

3.7 인터페이스

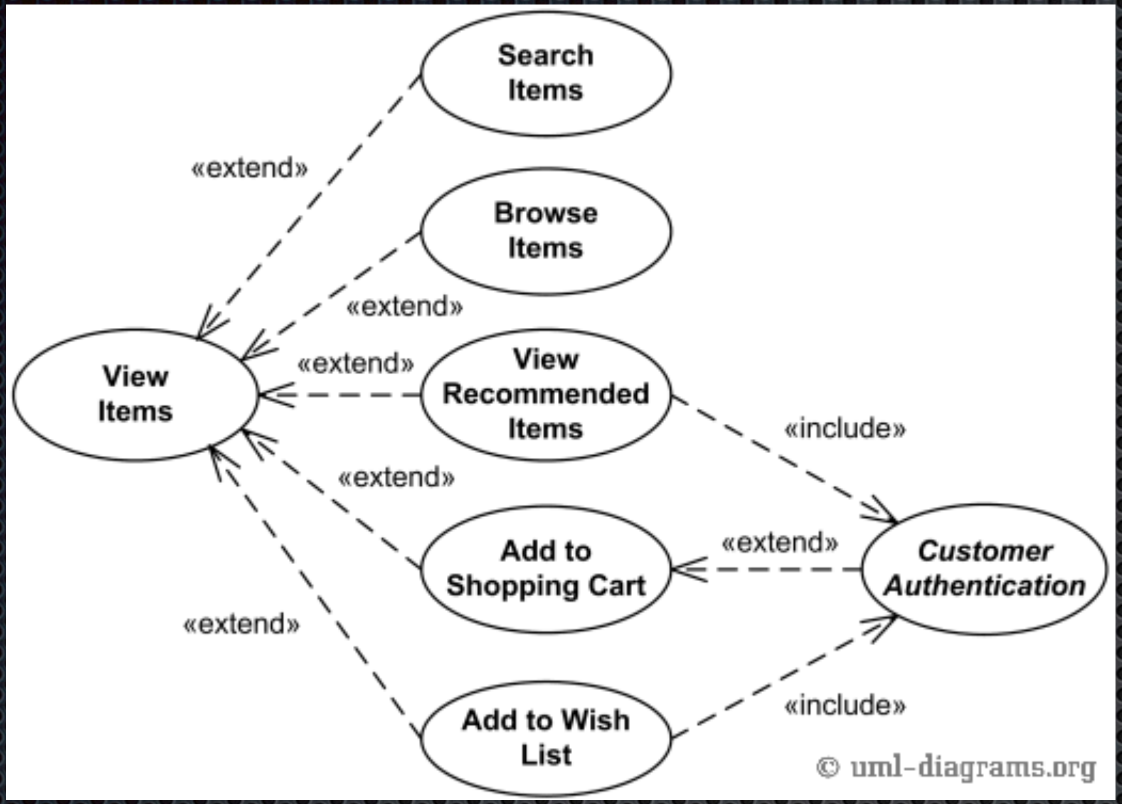
3.8 법적 제약사항

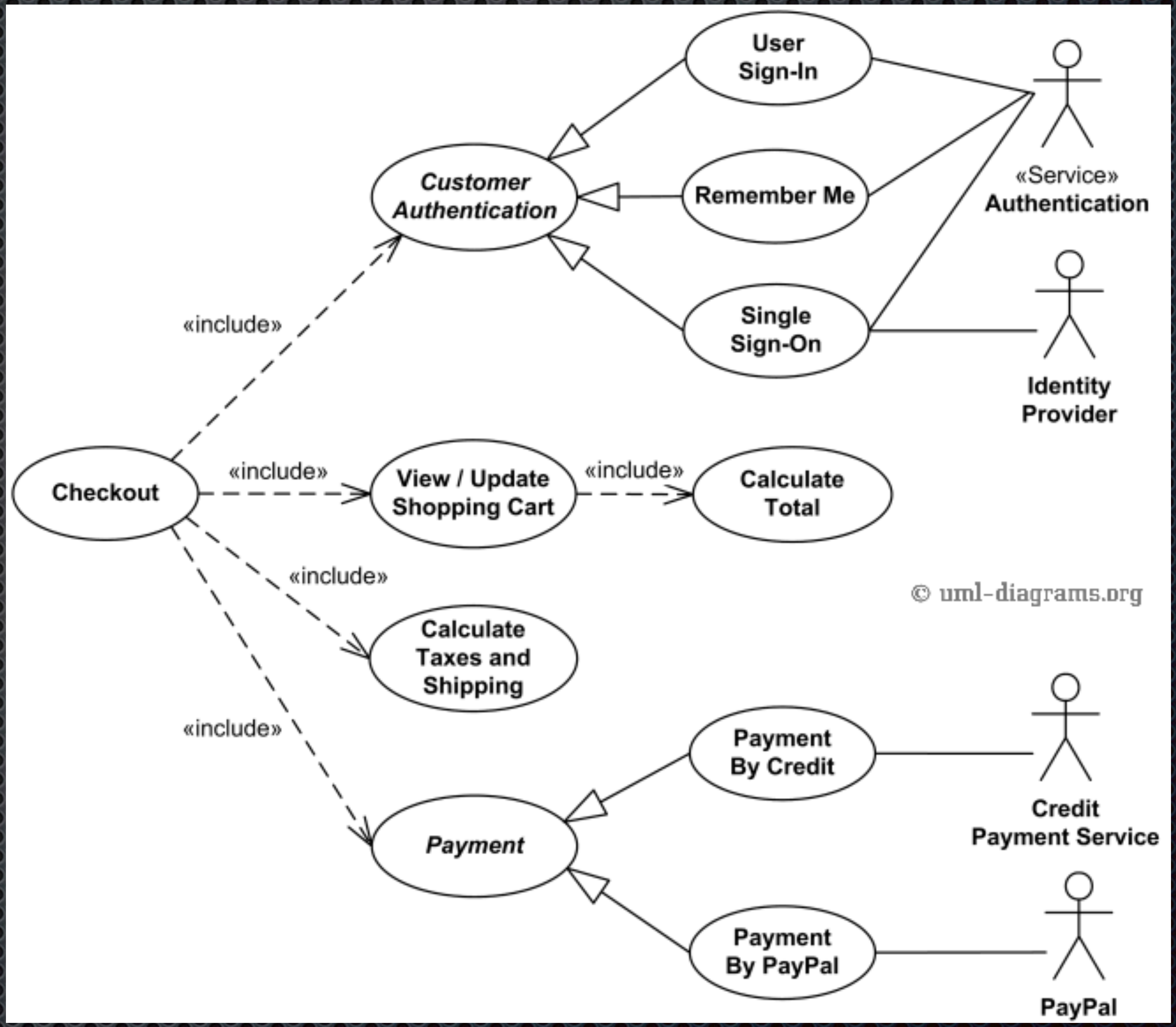
# Top Level Use Case Example




- 각각의 기능에 대한 설명

# Use Case Example

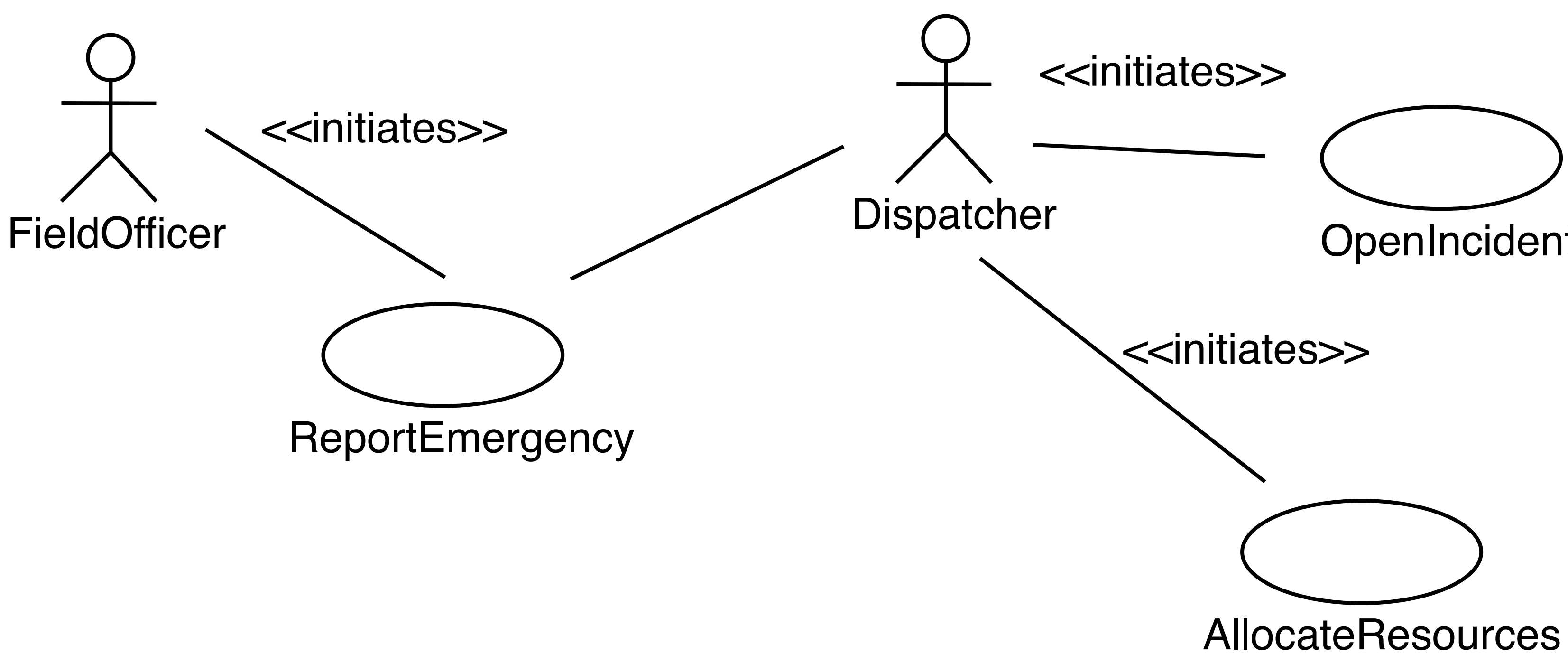


# Use Case Document

- Name of Use Case
- Actors
- Entry condition (optional)
- Flow of Events
- Exit condition (optional)
- Exceptions
- Special Requirements

# Example



<<initiates>>
<<initiates>>
FieldOfficer
Dispatcher
OpenIncident
<<initiates>>
ReportEmergency
AllocateResources

# Use case Example

- Use case name: ReportEmergency
- Participating Actors:
  - Field Officer (Bob and Alice in the Scenario)
  - Dispatcher (John in the Scenario)
- Exceptions:
  - The FieldOfficer is notified immediately if the connection between terminal and central is lost.
  - The Dispatcher is notified immediately if the connection between a FieldOfficer and central is lost.
- Flow of Events: on next slide.
- Special Requirements (비기능적 요구사항):
  - The FieldOfficer's report is acknowledged within 30 seconds. The selected response arrives no later than 30 seconds after it is sent by the Dispatcher.

# Use case Example

- The FieldOfficer activates the “Report Emergency” function of her terminal. The system responds by presenting a form to the officer.
- The FieldOfficer fills the form, by selecting the emergency level, type, location, and brief description of the situation. The FieldOfficer also describes a response to the emergency situation. Once the form is completed, the FieldOfficer submits the form, and the Dispatcher is notified.
- The Dispatcher creates an Incident in the database by invoking the OpenIncident use case. He selects a response and acknowledges the report.
- The FieldOfficer receives the acknowledgment and the selected response.

# 비기능적 요구사항

- Usability
- Reliability
- Performance
- Supportability
- Implementation
- Interface
- Operation
- Packaging
- Legal

# Career Paths for a SW/ Contents Developer

- As a solution programmer
- As a tool programmer
- As an appearance designer
  – Sound/Visual/…
- As a contents designer
  – Stories/…
- As a software designer
- As a technology developer
- As a producer

# What is the Producer?

- Difference the producer has
  - Not specialized to one aspect of the system
  - Requires wide range of specialty
- The producer
  - Primary job: to deliver the contents/SW as a completed project

# What is Required for a Producer?

# What is Required for a Producer?

- The development process is long and complex
- The development team consists of people with varying interests and backgrounds
- Should
  - Guide every process whenever necessary
  - Protect the team and the contents/SW
- To make it possible, one need to have
  - Communicate everyone
    - Give motivation, confidence and guidance
  - Do whatever

- The producer is NOT a manager
- He/she should deliver a result
- Manager may turn down a project but the producer should protect it

# Example of a Game Producer

- Role of a game producer
  - Decision making
    - Be forward thinking
    - Build consensus
    - Apply good decision-making skills
      - Good decision-making: good process to make decision
  - Developments
    - Actively contribute
    - Generate game-design documentation
    - Handle hardware manufacturers
    - Handle platform issues (middleware, hardware, …)

- Productions
  - Develop a pre-production and production plan
    - Executive summary
    - Creative design document
    - Technical design document
    - Risk-management plan
    - Schedule
    - Budget and financial requirements
  - Handle QA
  - Affect budget/Interact with upper management
  - Handle legal/contractual issues
  - Handle licensing and branding

- Management
  - Manage assets and other resources
  - Manage big teams
  - Manage Art/Audio/Technical processes
  - Hire/interview
- Public relations
  - Deliver animation
  - Handle PR
  - Help sales

– Rule of thumb

- Know games
- Lean new/old things
- Manage your time
- Take ownership

– Requirements

- Possess industry experience
- Provide clarity and focus
- Teach others when it is necessary

- Sow discipline
  - Set goals for people and encourage them
  - Obtain commitments from each team members
  - Measure progress and identify when the work can be done more efficiently
  - Hold others accountable for their actions and their commitments

# Software Production Methods

- Code-like-hell, Fix-like-hell
  - Well, most of you may do like this
  - Generally exhausts programmer. Prone to failure
  - Only acceptable for a very small projects

- Code-like-hell, Fix-like-hell
- Increments to Completion
  - Parallel progressing different modules
    - Increments to the already existing system/platform
  - Integrating at the final stage
  - High-level coordination is difficult

- Code-like-hell, Fix-like-hell
- Increments to Completion
- The cascade
  - Put almost every resources on one or a few stages
  - Quick development is possible
  - Not desirable when a design change is required

- Code-like-hell, Fix-like-hell
- Increments to Completion
- The cascade
- Iterate until you drop
  - Do the development iteration as many as possible
  - Adapting to change requirements
  - Weak when poor management is applied

# Planning and Scheduling

- Top-down approach
  - Good in theory
  - Could be bad when end-level requirements are not considered well
- Bottom-up approach
- Resource/Time-constrained Planning

- Resource/Time-constrained planning
  - Do not
    - Plan for overtime
    - Plan to hire additional personnel
    - Plan to work during holidays and vacations
    - Schedule work for the team leader (?)
  - Do
    - Make time for testing after a task is completed

# Summary

- Good producing
  - Comes from thorough understanding
    - Of the project
    - Of the project member
    - Of the design aspects required
    - Of the technical aspects required
  - Comes from determined decision on each and every aspect of the process
  - Comes from consensus by the project members